

1. A particle of mass m moving with velocity v collides with a stationary particle of mass $2m$. After collision, they stick together and continue to move together with velocity
- [2023 (10 Apr Shift 1)]
- (1) $\frac{v}{3}$
(2) $\frac{v}{4}$
(3) v
(4) $\frac{v}{2}$
2. An average force of 125 N is applied on a machine gun firing bullets each of mass 10 g at the speed of 250 m s^{-1} to keep it in position. The number of bullets fired per second by the machine gun is :
- [2023 (11 Apr Shift 1)]
- (1) 50
(2) 25
(3) 100
(4) 5
3. A nucleus disintegrates into two nuclear parts, in such a way that ratio of their nuclear sizes is $1 : 2^{1/3}$. Their respective speed have a ratio of $n : 1$. The value of n is _____
- [2023 (11 Apr Shift 2)]
4. Two bodies are having kinetic energies in the ratio $16 : 9$. If they have same linear momentum, the ratio of their masses respectively is:
- [2023 (13 Apr Shift 1)]
- (1) $3 : 4$
(2) $9 : 16$
(3) $16 : 9$
(4) $4 : 3$
5. A bullet of 10 g leaves the barrel of gun with a velocity of 600 m s^{-1} . If the barrel of gun is 50 cm long and mass of gun is 3 kg, then value of impulse supplied to the gun will be:
- [2023 (13 Apr Shift 1)]
- (1) 6 N s
(2) 3 N s
(3) 36 N s
(4) 12 N s

ANSWER KEYS

1. (1) 2. (1) 3. (2) 4. (2) 5. (1)

1. (1)

The law of conservation of momentum is given as

$$m_1 u + m_2 u' = (m_1 + m_2) v'$$

The given data is

$$u = v$$

$$u' = 0 \text{ m s}^{-1}$$

$$m_1 = m$$

$$m_2 = 2m$$

Using the conservation of momentum equation,

$$mv + 2m(0) = (m + 2m) v'$$

$$\Rightarrow v' = \frac{mv}{3m} = \frac{v}{3}$$

2. (1)

The formula to calculate the net momentum of the gun and bullet system can be written as

$$F = nmv \dots (1)$$

Substitute the values of the known parameters into equation (1) and solve to calculate the required number of bullets.

$$125 = n \times \frac{10}{1000} \times (250)$$

$$\Rightarrow n = 50$$

3. (2)

Using the law of Momentum conservation

$$m_1 v_1 + m_2 (-v_2) = 0$$

$$\Rightarrow m_1 v_1 = m_2 v_2$$

$$\Rightarrow \left(\frac{m_1}{m_2} \right) = \left(\frac{v_2}{v_1} \right)$$

$$\Rightarrow \frac{\frac{4\pi\rho r_1^3}{3}}{\frac{4\pi\rho r_2^3}{3}} = \left(\frac{v_2}{v_1} \right)$$

It is given

$$\frac{r_1}{r_2} = \frac{1}{2^{\frac{1}{3}}}$$

$$\text{and } \left(\frac{r_1}{r_2} \right)^3 = \left(\frac{v_2}{v_1} \right)$$

Therefore,

$$\left(\frac{1}{2} \right) = \left(\frac{v_2}{v_1} \right)$$

$$\Rightarrow \frac{v_1}{v_2} = \frac{2}{1}$$

4. (2)

The kinetic energy and the momentum of a moving object are related by

$$K = \frac{p^2}{2m} \dots (1)$$

Hence, the ratio of the kinetic energies of two objects having masses m and m' can be calculated as follows:

$$\frac{K}{K'} = \frac{\frac{p^2}{2m}}{\frac{p'^2}{2m'}}$$

$$= \frac{m'}{m} \dots (2)$$

Substitute the ratio of the kinetic energies into equation (2) and simplify to obtain the ratio of their masses.

$$\Rightarrow \frac{16}{9} = \frac{m'}{m}$$

$$\Rightarrow m : m' = 9 : 16$$

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The formula to calculate the impulse (J) on the gun is given by

$$J = m(v - u) \quad \dots (1)$$

where, m is the mass, u is the initial velocity and v is the final velocity of the bullet.

Substitute the values of the known parameters into equation (1) to calculate the required impulse.

$$J = 10 \text{ g} \times \frac{1 \text{ kg}}{1000 \text{ g}} \times (600 - 0) \text{ m s}^{-1} \\ = 6 \text{ N s}$$